

Requirements:

1. Implement standard technical indicators as a baseline:
 - RSI (Relative Strength Index)
 - Bollinger Bands
 - MACD (Moving Average Convergence Divergence)
 - Moving Averages (SMA and EMA)
2. Create at least 3 custom technical indicators such as:
 - A modified RSI that uses different weighting for up and down days
 - A volatility-adjusted Bollinger Band that changes width based on market regime
 - A multi-timeframe momentum indicator that combines signals from different periods
 - An adaptive moving average that adjusts its period based on volatility
 - A volume-weighted indicator that incorporates trading volume
3. Develop a signal generation framework that:
 - Creates clear buy/sell signals for both standard and custom indicators
 - Handles signal conflicts (when different indicators suggest opposite actions)
 - Incorporates signal strength (how strongly an indicator suggests a trade)
 - Implements filters to reduce false signals
4. Build a backtesting system that:
 - Tests each indicator individually on at least 5 different stocks
 - Compares the performance of standard vs. custom indicators